

Intravitreal Bevacizumab at 4-Month Intervals for Prevention of Macular Edema after Plaque Radiotherapy of Uveal Melanoma

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Purpose: To evaluate the efficacy of intravitreal bevacizumab for prevention of macular edema after plaque radiotherapy of uveal melanoma.

Design: Retrospective, single-center, nonrandomized, interventional comparative study.

Participants: Patients with uveal melanoma treated with plaque radiotherapy were divided into 2 groups: a bevacizumab group and a control group.

Intervention: The bevacizumab group received intravitreal bevacizumab injection at the time of plaque removal and every 4 months thereafter for 2 years (total, 7 injections). The control group had no intravitreal bevacizumab injection. Both groups had periodic follow-up with ophthalmoscopy and optical coherence tomography (OCT).

Main Outcome Measures: Development of OCT-evident macular edema.

Results: There were 292 patients in the bevacizumab group and 126 in the control group. The median foveolar radiation dose was 4292 cGy (bevacizumab) and 4038 cGy (control; $P = 0.327$). The cumulative incidence of OCT-evident macular edema over 2 years (bevacizumab group vs. control group) was 26% versus 40% ($P = 0.004$), respectively; that for clinically evident radiation maculopathy was 16% versus 31% ($P = 0.001$), respectively; that for moderate vision loss was 33% versus 57% ($P < 0.001$), respectively; and that for poor visual acuity was 15% versus 28% ($P = 0.004$), respectively. There was no statistically significant difference in clinically evident radiation papillopathy ($P = 0.422$). Kaplan-Meier estimates at 2 years showed statistically significantly reduced rates of OCT-evident macular edema ($P = 0.045$) and clinically evident radiation maculopathy ($P = 0.040$) in the bevacizumab group compared with controls.

Conclusions: Patients receiving intravitreal bevacizumab injection every 4 months after plaque radiotherapy for uveal melanoma demonstrated OCT-evident macular edema, clinically evident radiation maculopathy, moderate vision loss, and poor visual acuity less frequently over a period of 2 years than patients not receiving the injections. *Ophthalmology* 2014;121:269-275 © 2014 by the American Academy of Ophthalmology.



Radiation maculopathy is a potentially vision-threatening complication after irradiation of the eye.¹ It manifests as macular edema, retinal hemorrhage and exudation, nerve fiber layer infarction, and perivascular sheathing, and can progress to retinal or optic disc neovascularization with potential for vitreous hemorrhage, loss of vision, neovascular glaucoma, and loss of the eye.¹ In a study of 1106 patients with uveal melanoma treated with plaque radiotherapy, Shields et al² reported that visual acuity declined to 20/200 or worse in 34% at 5 years and 68% at 10 years after therapy. Similar findings were reported by the Collaborative Ocular Melanoma Study group in an analysis of 620 irradiated eyes with uveal melanoma in which visual acuity declined to 20/200 or worse in 43% at 3 years after plaque radiotherapy.³ Most of these poor

visual outcomes are attributable to radiation maculopathy, radiation retinopathy, radiation papillopathy, and their sequelae. Over a median follow-up of 2 years, Horgan et al⁴ found that visual loss occurred significantly more frequently in postradiotherapy eyes with optical coherence tomography (OCT)-evident macular edema than those without it. As is evident, radiation-related macular edema is one of the major causes of vision loss after treatment of uveal melanoma.

The presence of uveal melanoma can lead to elevated vitreous levels of vascular endothelial growth factor (VEGF).⁵ Radiotherapy can increase VEGF production further, driven by ischemic factors.⁵ Direct tumoricidal effects of radiation are mediated through free radical formation, damage to the cellular structures and DNA,